

MengKu (Nick) Chen

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EDUCATION

Brandeis University

Master's Degree, Computer Science

Waltham, MA

Graduation Date: May 2023

National Taiwan Ocean University

Bachelor's Degree, Bioscience and Biotechnology

Keelung, Taiwan

Graduation Date: Jun 2019

SKILLS

- Languages: **Java**, **Python**, **JavaScript**, **TypeScript**, **SQL**, HTML/CSS, PHP, Ruby
- Frameworks: **Spring MVC**, **Spring Boot**, **Spring Cloud**, **Hibernate**, **Node**, **Express**, **React**, **Vue**, Flask, Django, LangChain, FastAPI, React Native
- Databases & Cloud Platforms: **MySQL**, **MongoDB**, SQL Sever, DynamoDB, Redis, PostgreSQL, AWS EC2, AWS RDS, Azure, Google Cloud Platform, S3.
- Tools & OS: **Docker**, **K8s**, **CI/CD**, **RabbitMQ**, **Linux**, Agile, Git, Postman, Pytest, JUnit, Jenkins, Bitbucket, Apache Tomcat, Kafka.
- Certifications: **AWS Certified Solutions Architect**

WORK EXPERIENCE

NNRoad

Software Engineer

San Jose, CA

Sep 2024 – Current

- Deployed a GenAI agent leveraging **LangChain**, **RAG**, and the **OpenAI API**, integrating vector databases (**FAISS**) for context retrieval and asynchronous pipelines for high throughput. Improved policy update velocity by 45%.
- Built a Human Resources system with **TypeScript**, **Vue**, **Java Spring Boot**, **MyBatis**, and **RabbitMQ**, implementing event-driven architecture, role-based access control (**RBAC**), and schema-driven dynamic UI configuration, which boosted operational flexibility by 40%.
- Engineered a payroll management platform using **Node.js**, **React**, and **MongoDB**, applying microservices principles and caching layers (**Redis**), automating payroll workflows to reduce manual processing time by 60%.
- Established **CI/CD** pipelines with **GitHub Actions**, containerized deployments with **Docker** and **Kubernetes** orchestration.
- Mentored 5 interns on architecture for AI chatbots, blog generators, and attendance systems, instilling best practices in clean code and accelerating project delivery by 30%.

Axim Solutions

Junior Software Engineer

Richmond, VA

Sep 2023 – June 2024

- Adhered to Agile methodologies throughout the software development life cycle (SDLC) for the web-based CMS project utilizing the **Linux**, **Apache**, **MySQL**, **PHP** (LAMP), leading to a 15% decrease in project timeline.
- Set up request rate limiting with **AWS API Gateway** and combined it with **Redis** caching to handle high request loads, improving system stability and scalability.
- Switched from HTTP requests initiated by the frontend to **WebSocket** for real-time updates in message pushing.
- Introduced **RabbitMQ** for asynchronous content creation task processing, increasing task handling from 200 to 600 per hour and reducing content creation request response time from 3 seconds to 1 second.
- Managed the migration of over 40 old websites to **AWS EC2**, utilizing **Uptime** and **AWS CloudWatch** monitoring tools to reduce downtime from 5 hours to 4 hours per month.

Telesign | Leading Digital Identity Company

Software Engineering Intern

Los Angeles, CA

Jun 2022 - Aug 2022

- Collaborated with cross-functional teams to launch a new product, integrated **FastAPI** for backend services, and automated customer information updates using **Django** and **Pytest**.
- Monitored **SQLServer** to identify slow queries, used execution plans to detect full table scans, created proper indexes, and reduced the execution time of thousand-row stored procedures from 2.5 seconds to 1.5 seconds.
- Integrated **skeleton screens** to improve perceived performance during slow network conditions, displaying placeholder content during data loading and enhancing user experience.
- Wrapped virtual scrolling with **react-virtualized** to render only elements in the viewport, solving the problem of rendering long lists (2000+ items) and preventing lag.

Double Information Technology

Full Stack Engineer

Taipei, Taiwan

Sep 2020 - Jun 2021

- Developed a resume management system based on **Java**, **Spring Boot**, **Spring Cloud**, **Python**, **Flask**, and **Vue**, supporting user creation and parsing of resumes, HR management and search, and generation of PDF and Word files.
- Utilized **DevTools** to identify frontend performance issues and implemented image lazy loading, **splitChunks** code splitting, and **GZIP** compression, reducing initial load time from 5300ms to 860ms.
- Led the design of the permission control scheme, implementing user menu and operation permissions management, and applied it to multiple subsequent business systems, significantly accelerating the infrastructure setup of new systems.
- Used **CompletableFuture** to handle parsing and saving processes asynchronously, allowing the main thread to only wait for the parsing task to start. Parsing and saving were executed in parallel, reducing response time to 1.5 seconds.